



V93000 Specifications

V93000 Wave Scale PS Card

Version 8.6.0

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Notices

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Wave Scale PS Card

The Wave Scale Precision Source (WSPS) card can be connected in the following two ways:

The "WSPS card using N2601P interconnect cable (Topic 255027)" is used to extend the capabilities of the WSRF card for applications that require extremely good phase noise performance.

The "WSPS card using N2601PD direct pogo cable (WSPS-PD) (Topic 344348)" connects directly to the DUT-IF to provide high precision converter clocks and ADC signals.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

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Wave Scale PS card, using N2601P interconnect cable

The Wave Scale Precision Source card (WSPS) using N2601P interconnect cable is used to extend the capabilities of the Wave Scale RF (WSRF) card for applications that require extremely good phase noise performance.

The WSPS-8 card using N2601P interconnect cable is used to extend the capabilities of the WSRF-8 card for applications that require extremely good phase noise performance.

For information on the WSPS card, which connects directly to the DUT-IF to provide high precision converter clocks and ADC signals, refer to the section "Wave Scale Precision Source card, using N2601PD direct pogo cable (Topic 344348)".

Wave Scale PS stimulus

For the combined WSPS + WSRF stimulus specifications, refer to the section "Wave Scale RF stimulus (Topic 252876)". The specifications are also repeated within this section for completeness.

Wave Scale PS stimulus random jitter

For the combined WSRF + WSPS stimulus random jitter specifications, refer to the section "Wave Scale RF stimulus random jitter (Topic 252941)". The specifications are also repeated below for completeness.

Integration bandwidth : 10kHz to 10MHz

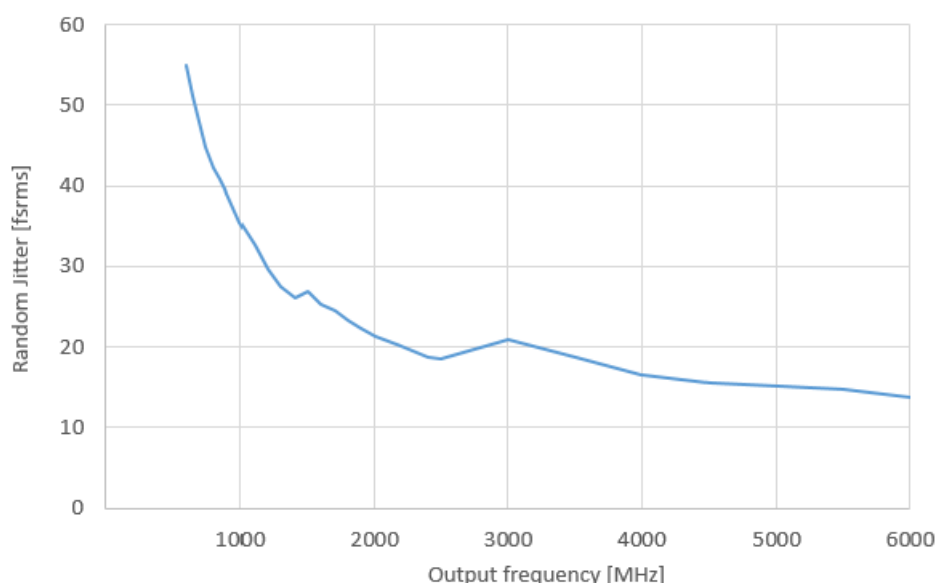
WSRF + WSPS stimulus random jitter max

Frequency [MHz]	Random jitter max [$f_{s_{rms}}$]
600 $\leq f < 800$	75
800 $\leq f < 1000$	59
1000 $\leq f < 1400$	46
1400 $\leq f < 1600$	37
1600 $\leq f < 2000$	36
2000 $\leq f < 4000$	32
4000 $\leq f < 5000$	26
5000 $\leq f \leq 6000$	22

WSRF + WSPS stimulus random jitter typ

Frequency [MHz]	Random jitter typ [$f_{s_{rms}}$]
600	56
885	40
1900	23
2400	19
5800	15

The following figure shows the typical measured stimulus random jitter as a function of output carrier frequency.



Wave Scale PS stimulus phase noise

For the combined WSRF + WSPS stimulus phase noise specifications, refer to the section "Wave Scale RF stimulus phase noise (Topic 252936)". The specifications are also repeated below for completeness.

WSRF + WSPS stimulus phase noise max [dBc/Hz]

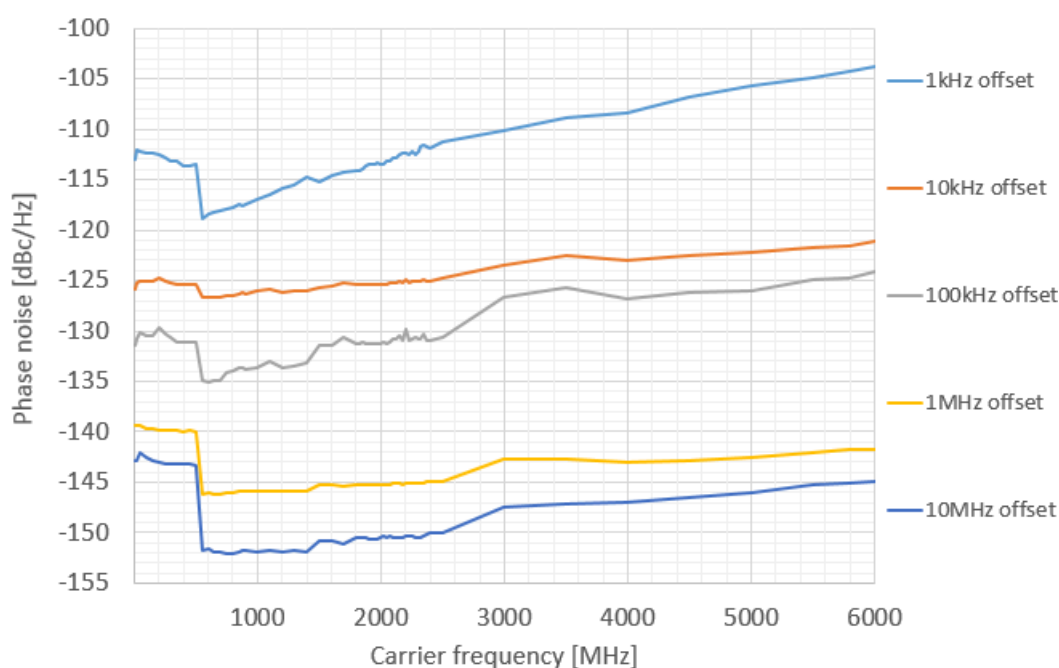
Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-104	-119	-124	-132	-134
320	-104	-119	-124	-132	-134
545	-112	-121	-129	-142	-148
600	-112	-121	-129	-142	-148
885	-111	-120	-128	-142	-148
1900	-108	-119	-125	-141	-147
2400	-105	-119	-125	-141	-147
5800	-95	-115	-117	-138	-141

WSRF + WSPS stimulus phase noise typ [dBc/Hz]

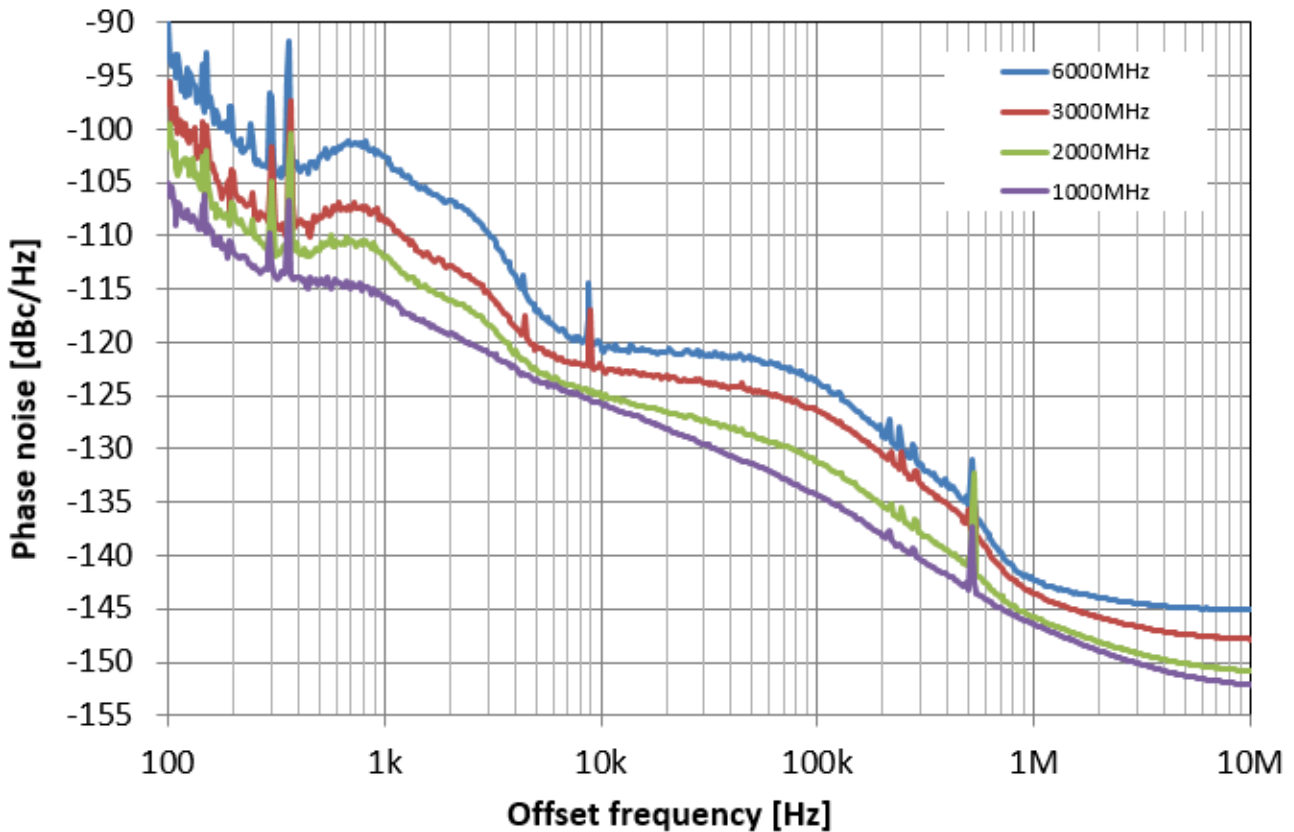
Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-111	-124	-130	-139	-141
320	-111	-125	-130	-139	-143
545	-118	-126	-134	-145	-151

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
600	-117	-126	-134	-145	-151
885	-116	-125	-133	-145	-151
1900	-112	-125	-131	-144	-150
2400	-110	-124	-130	-144	-149
5800	-103	-121	-124	-141	-144

The following graph for WSRF+WSPS shows the typical measured stimulus phase noise versus carrier frequency for several offset frequencies.



The following graph for WSRF+WSPS shows the typical measured stimulus phase noise versus offset frequency for several carrier frequencies.



Wave Scale PS stimulus EVM (error vector magnitude)

For the combined WSRF + WSPS stimulus EVM specifications, refer to the section "Wave Scale RF stimulus EVM (Topic 252937)". The specifications are also repeated below for completeness.

WSRF + WSPS stimulus EVM

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM max	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.5%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB , using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-15 dBm to -2 dBm)	0.55%	0.45%
5.6 (-15 dBm to -2 dBm)	0.55%	0.47%

Optimizer: balanced, NoiseBandwidth=80 MHz

802.11ac 160MHz

Frequency [GHz]	EVM
5.2 (-15 dBm to -2 dBm)	0.7% (max)
5.2 (-11 dBm to -2 dBm)	0.54% (typ)
5.2 (-15 dBm to -11 dBm)	0.62% (typ)
5.6 (-15 dBm to -2 dBm)	0.75% (max)
5.6 (-11 dBm to -2 dBm)	0.58% (typ)
5.6 (-15 dBm to -11 dBm)	0.66% (typ)

Optimizer: balanced, NoiseBandwidth=40 MHz

Wave Scale PS stimulus ACLR (adjacent channel leakage ratio)

For the combined WSRF + WSPS stimulus ACLR specifications, refer to the section "Wave Scale RF stimulus ACLR (Topic 252938)". The specifications are also repeated below for completeness.

WSRF + WSPS stimulus ACLR

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	ACLR [dBc]	ACLR2 [dBc]
2 (-30 dBm to +4 dBm)	-68 (typ)	-72.5 (typ)
2 (-25 dBm to +1 dBm)	-66.5 (max)	-71.5 (max)

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 6dB

Wave Scale PS stimulus spurs

For the combined WSRF + WSPS stimulus spur specifications, refer to the section "Wave Scale RF stimulus spurs (Topic 255613)". The specifications are also repeated below for completeness.

Measured at 0dBm stimulus power

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

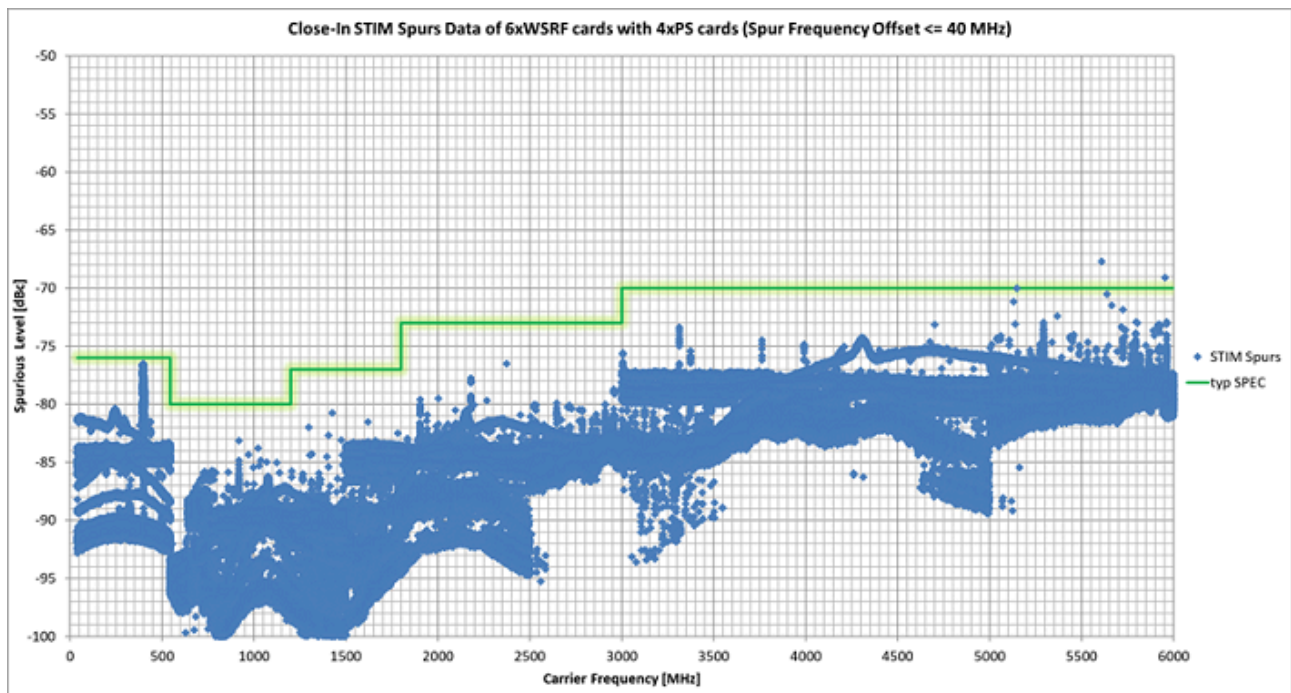
WSRF + WSPS stimulus spurs max

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.01	6	-63

WSRF + WSPS stimulus spurs typ

Frequency from [GHz]	Frequency to [GHz]	Typ spur [dBc]
0.04	0.55	-76
0.55	1.2	-80
1.2	1.8	-77

Frequency from [GHz]	Frequency to [GHz]	Typ spur [dBc]
1.8	3	-73
3	6	-70



WSPS stimulus nominal spurious performance

Valid for spur frequency offset above +/- 40MHz

WSRF + WSPS stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	6	-63

Wave Scale PS-8 stimulus

For the combined WSPS-8 + WSRF-8 stimulus specifications, refer to the section "Wave Scale RF-8 with RFIM-8 stimulus (Topic 354043)". The specifications are also repeated within this section for completeness.

Wave Scale PS-8 stimulus random jitter

For the combined WSRF-8 + WSPS-8 stimulus random jitter specifications, refer to the section "Wave Scale RF-8 stimulus random jitter (Topic 354352)". The specifications are also repeated below for completeness.

Integration bandwidth : 10kHz to 10MHz

WSRF-8 + WSPS-8 stimulus random jitter max

Frequency [MHz]	Random jitter max [fs_{rms}]
$600 \leq f < 800$	83
$800 \leq f < 1000$	64

Frequency [MHz]	Random jitter max [$f_{s_{rms}}$]
1000 $\leq f < 1400$	53
1400 $\leq f < 1600$	40
1600 $\leq f < 2000$	36
2000 $\leq f < 2500$	33
2500 $\leq f < 4000$	31
4000 $\leq f < 5000$	24
5000 $\leq f < 6000$	21
6000 $< f \leq 7000$	33

WSRF-8 + WSPS-8 stimulus random jitter typ

Frequency [MHz]	Random jitter typ [$f_{s_{rms}}$]
600	54
885	41
1900	23
2400	19
5800	15
7000	18

Wave Scale PS-8 stimulus phase noise

For the combined WSRF-8 + WSPS-8 stimulus phase noise specifications, refer to the section "Wave Scale RF-8 stimulus phase noise (Topic 354349)". The specifications are also repeated below for completeness.

WSRF-8 + WSPS-8 stimulus phase noise [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-104	-119	-124	-132	-134
320	-104	-119	-124	-132	-134
545	-111	-120	-129	-142	-148
600	-111	-120	-129	-142	-148
885	-110	-120	-128	-142	-148
1900	-107	-119	-125	-141	-147
2400	-105	-118	-124	-141	-147
5800	-94	-115	-117	-138	-141

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
7000	-94	-112	-116	-135	-139

WSRF-8 + WSPS-8 stimulus phase noise typ [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-113.5	-124.9	-131.3	-139.7	-142.4
320	-113.5	-124.9	-131.1	-140.1	-143.3
545	-116.9	-126.2	-136.7	-146.9	-152.6
600	-116.9	-126.3	-136.8	-146.9	-152.5
885	-116.1	-125.5	-135	-146.3	-152
1900	-113.9	-124.9	-132	-145.7	-150.9
2400	-112.4	-124.5	-131.5	-145.5	-150.4
5800	-106.3	-119.4	-124.6	-141.7	-145.2
7000	-104.5	-117.7	-121.2	-139.5	-143.2

Wave Scale PS-8 stimulus EVM (error vector magnitude)

For the combined WSRF-8 + WSPS-8 stimulus EVM specifications, refer to the section "Wave Scale RF-8 stimulus EVM (Topic 354693)". The specifications are also repeated below for completeness.

WSRF-8 + WSPS-8 stimulus EVM

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM max	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.52%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB , using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-15 dBm to -2 dBm)	0.55%	0.4%
5.6 (-15 dBm to -2 dBm)	0.55%	0.42%

Optimizer: balanced, NoiseBandwidth=20 MHz

802.11ac 160MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-11 dBm to -2 dBm)	0.7%	0.53%
5.2 (-15 dBm to -11 dBm)	0.7%	0.54%
5.6 (-11 dBm to -2 dBm)	0.75%	0.56%
5.6 (-15 dBm to -11 dBm)	0.75%	0.57%

Optimizer: balanced, NoiseBandwidth=10 MHz

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-33 to < -30	0.81%	-41.83
	-30 to < -25	0.59%	-44.58
	-25 to < -16	0.46%	-46.74
	-16 to < -7	0.40%	-47.96
	-7 to < -2	0.45%	-46.94
	-2 to 0	0.54%	-45.35
	0 < to 3	0.72%	-42.85

Optimizer: balanced, NoiseBandwidth=9 MHz

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-30 to < -25	0.83%	-41.62
	-25 to < -18	0.66%	-43.61
	-18 to -5	0.61%	-44.29
	-5 < to < -2	0.65%	-43.74
	-2 to 0	0.68%	-43.35
	0 < to 3	0.86%	-41.31

Optimizer: balanced, NoiseBandwidth=20 MHz

Wave Scale PS-8 stimulus ACLR (adjacent channel leakage ratio)

For the combined WSRF + WSPS stimulus ACLR specifications, refer to the section "Wave Scale RF-8 stimulus ACLR (adjacent channel leakage ratio) (Topic 354573)". The specifications are also repeated below for completeness.

WSRF-8 + WSPS-8 stimulus ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, Center frequency: 2 GHz

Power level	ACLR [dBc]	ACLR2 [dBc]
-30 dBm to +4 dBm	-67.6 (typ)	-72.7 (typ)
-25 dBm to +1 dBm	-66.5 (max)	-71.5 (max)

Optimizer: balanced, NoiseBandwidth= 60 MHz, baseband attenuation 6 dB

Wave Scale PS-8 stimulus spurs

Measured at 0dBm stimulus power

For the combined WSRF-8 + WSPS-8 stimulus spur specifications, refer to the section "Wave Scale RF-8 stimulus spurs (Topic 354597)". The specifications are also repeated below for completeness.

Valid for spur frequency offset ranging from +/-10kHz to +/-100MHz

WSRF-8 + WSPS-8 stimulus spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.04	0.55	-76
0.55	1.2	-80
1.2	1.8	-77
1.8	3	-73
3	6	-70
6	8	-68

Valid for spur frequency offset above +/-40MHz

WSRF-8 + WSPS-8 stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	6	-63
6	7	-63
7	7.4	-60
7.4	7.8	-54
7.8	8	-48

Wave Scale PS measure

For the combined WSPS + WSRF measure specifications, refer to the section "Wave Scale RF measure (Topic 252877)". The specifications are also repeated within this section for completeness.

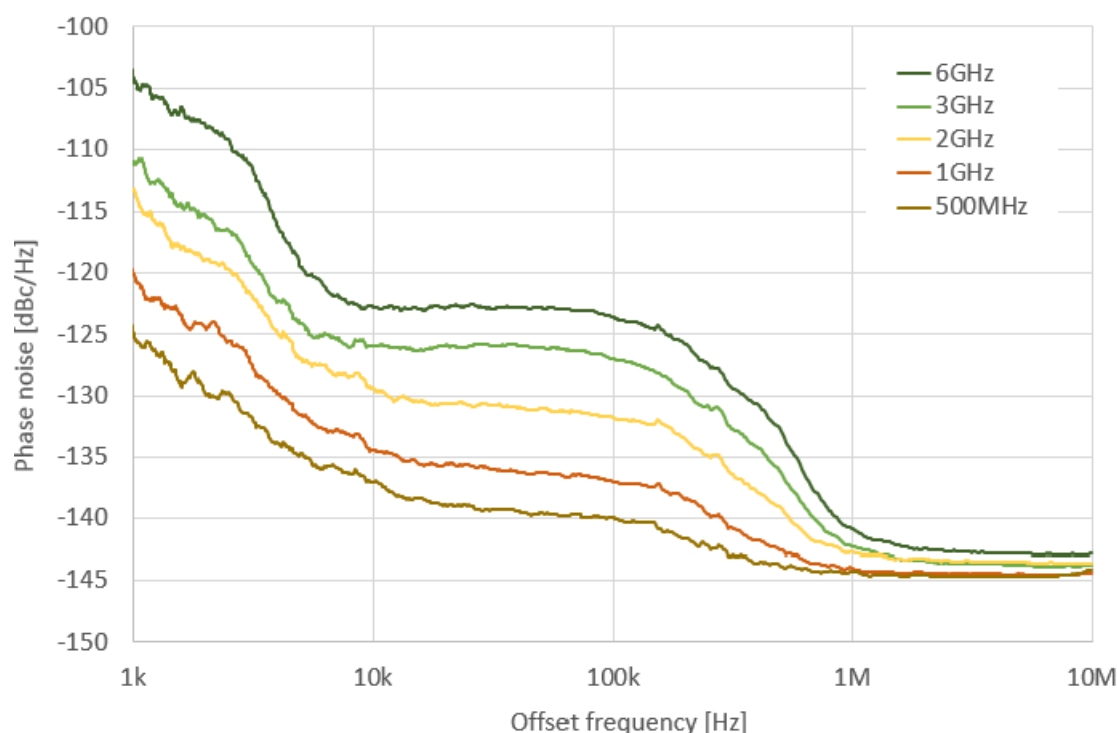
Wave Scale PS measure phase noise

For the combined WSRF + WSPS measure phase noise specifications, refer to the section "Wave Scale RF measure phase noise (Topic 344344)". The specifications are also repeated below for completeness.

WSRF + WSPS measured phase noise (measured with E8257D UNY option / Keysight Technologies) [dBc/Hz]

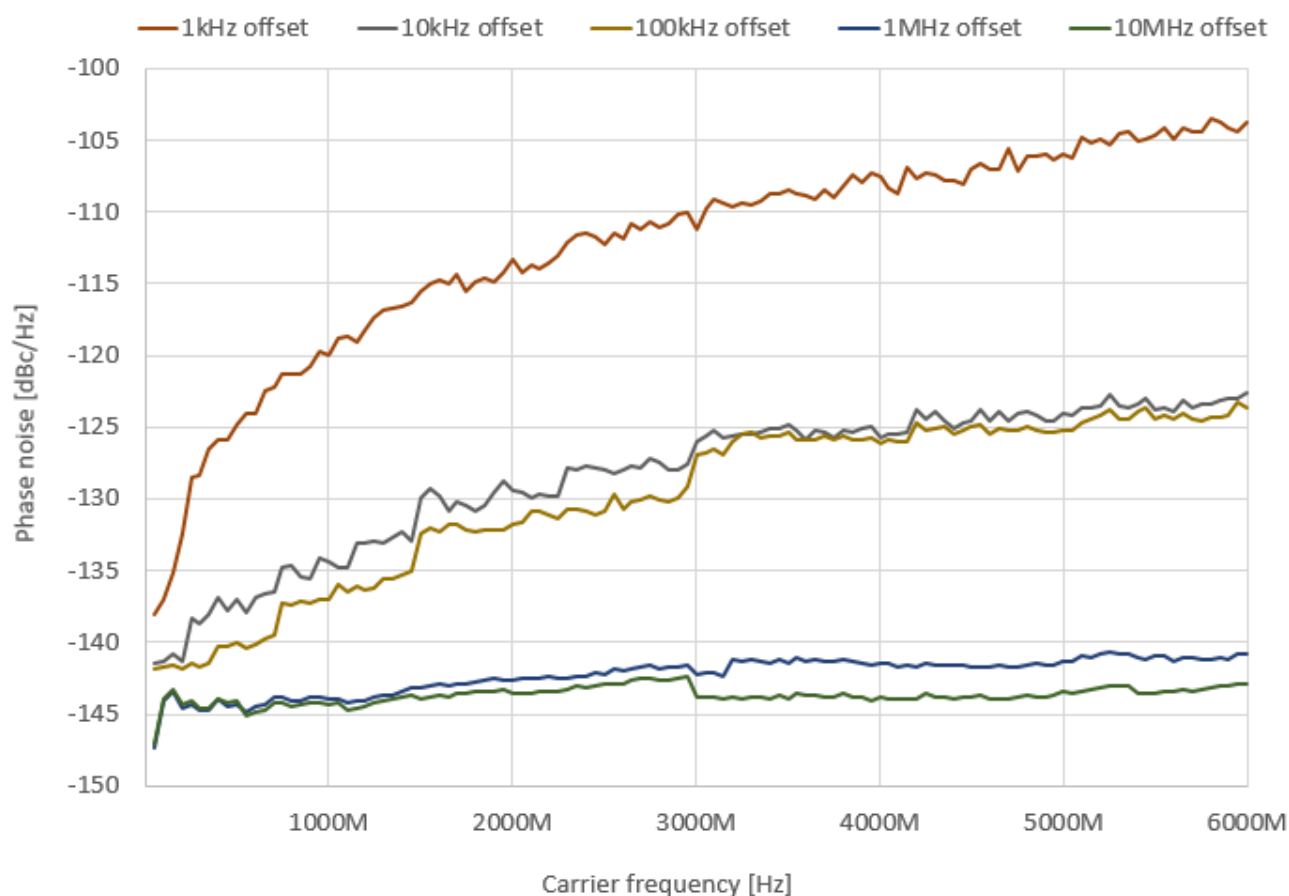
Input carrier frequency [GHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
0.6	-124	-137	-140	-145	-145
0.885	-120	-135	-137	-144	-144
1.9	-115	-130	-132	-143	-144
2.4	-112	-128	-131	-142	-143
5.8	-104	-123	-124	-141	-143

The following graph shows the typical WSRF + WSPS measured phase noise versus offset frequency for several input carrier frequencies.



Wave Scale PS measured phase noise (measured with E8257D UNY option / Keysight Technologies)

The following graph shows the typical WSRF + WSPS measured phase noise versus input carrier frequency for several offset frequencies.



WS PS measured phase noise performance

Wave Scale PS measure random jitter

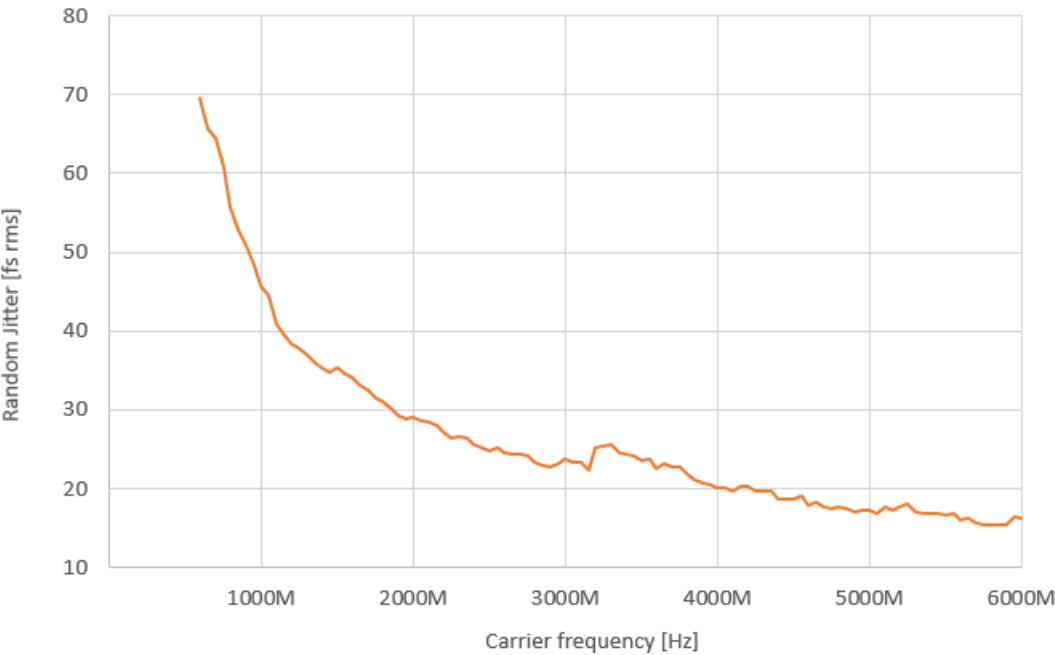
For the combined WSRF + WSPS measure random jitter specifications, refer to the section "Wave Scale RF measure random jitter (Topic 344347)". The specifications are also repeated below for completeness.

Integration bandwidth : 10kHz-10MHz

WSRF + WSPS measured random jitter (measured with E8257D UNY option / Keysight Technologies)

Input carrier frequency [GHz]	Random jitter [fs_{rms}]
0.6	69
0.885	52
1.9	29
2.4	25
5.8	15

The following figure shows the typical measured random jitter as a function of carrier frequency.



Measured random jitter (measured with E8257D UNY option / Keysight Technologies)

Wave Scale PS measure EVM (error vector magnitude)

For the combined WSRF + WSPS measure EVM specifications, refer to the section "Wave Scale RF measure EVM (Topic 252947)". The specifications are also repeated below for completeness.

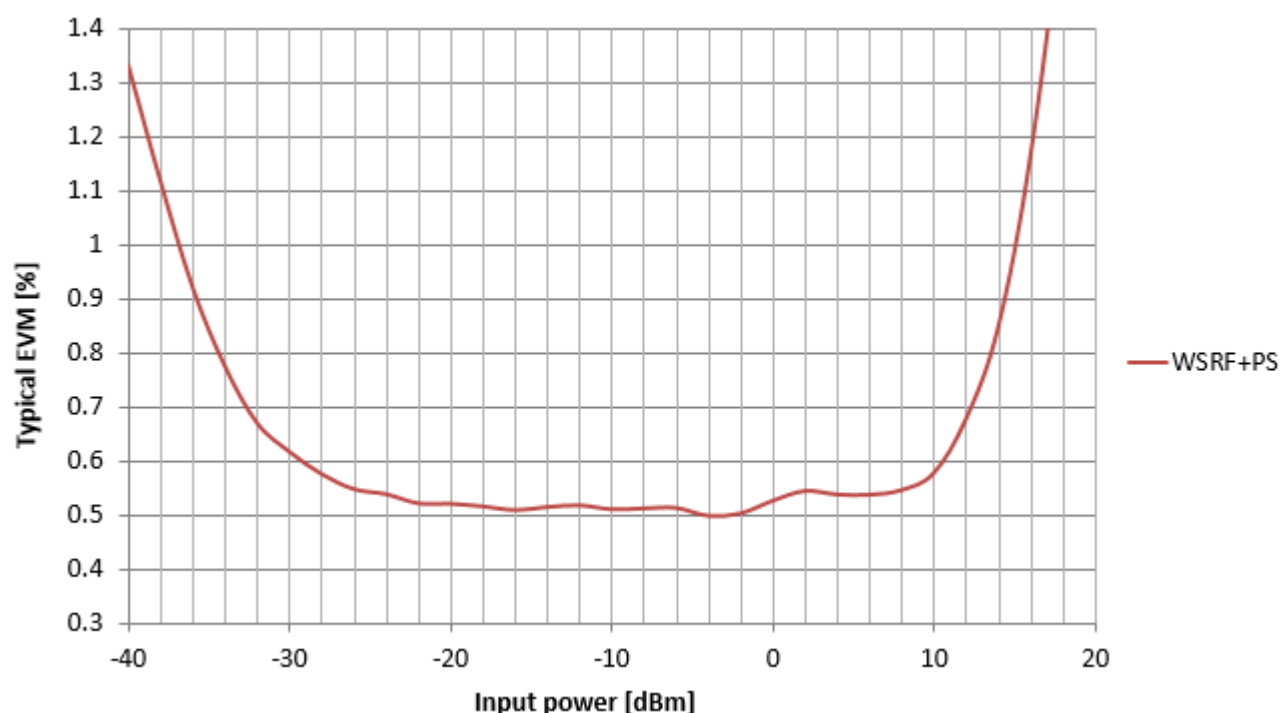
WSRF + WSPS measure EVM

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM
2 (-40 dBm to +2 dBm)	0.8% (max)
2 (-40 dBm to +14 dBm)	0.54% (typ)

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM
5.2 (-26 dBm to + 2 dBm)	1.0% (max)
5.2 (-26 dBm to +8 dBm)	0.54% (typ)



Wave Scale PS measure typ EVM 802.11ac 160MHz

Wave Scale PS measure ACLR (adjacent channel leakage ratio)

For the combined WSRF + WSPS measure phase ACLR specifications, refer to the section "Wave Scale RF measure ACLR (Topic 253378)". The specifications are also repeated below for completeness.

WSRF + WSPS measure ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

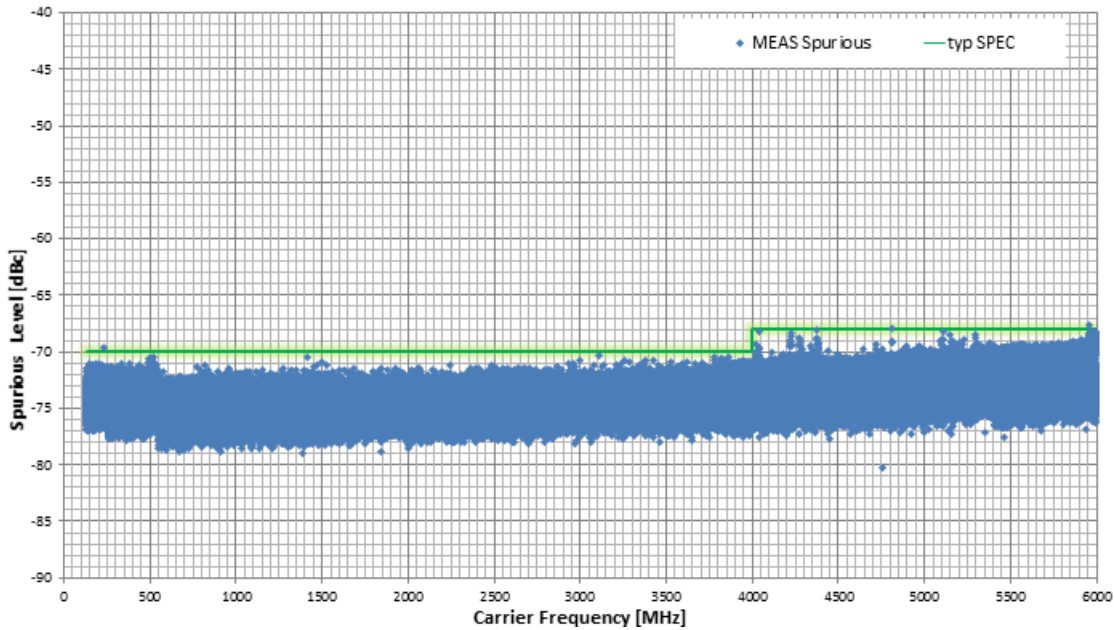
Frequency [GHz]	ACLR [dBc]
2 (-16 dBm to +6 dBm)	-67 (max)
2 (-16 dBm to +14 dBm)	-69 (typ)

Wave Scale PS measure spurs

For the combined WSRF + WSPS measure spur specifications, refer to the section "Wave Scale RF measure spurs (Topic 255614)". The specifications are also repeated below for completeness.

WSRF + WSPS measure spurs

Input frequency from [GHz]	Input frequency to [GHz]	Typ spurs level (<0.01%) [dBc]
0.13	4	-70
4	6	-68



WS PS measure typical spurious performance

Wave Scale PS-8 measure

For the combined WSPS-8 + WSRF-8 measure specifications, refer to the section "Wave Scale RF-8 with RFIM-8 measure (Topic 354044)". The specifications are also repeated within this section for completeness.

Wave Scale PS-8 measure EVM (error vector magnitude)

For the combined WSRF-8 + WSPS-8 measure EVM specifications, refer to the section "Wave Scale RF-8 measure EVM (Topic 354717)". The specifications are also repeated below for completeness.

WSRF-8 + WSPS-8 measure EVM

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM (max)	EVM (typ)
2 (-40 dBm to +10 dBm)	0.8%	0.47%

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM (max)	EVM (typ)
5.2 (-16 dBm to + 2 dBm)	1.0%	0.52%
5.2 (-26 dBm to -16 dBm)	1.0%	0.58%

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-33 to < -30	0.81%	-41.83
	-30 to < -25	0.59%	-44.58

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
	-25 to < -16	0.46%	-46.74
	-16 to < -7	0.40%	-47.96
	-7 to < -2	0.45%	-46.94
	-2 to 0	0.54%	-45.35
	0 < to 3	0.72%	-42.85

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-30 to < -25	0.83%	-41.62
	-25 to < -18	0.66%	-43.61
	-18 to -5	0.61%	-44.29
	-5 < to < -2	0.65%	-43.74
	-2 to 0	0.68%	-43.35
	0 < to 3	0.86%	-41.31

Wave Scale PS-8 measure ACLR (adjacent channel leakage ratio)

For the combined WSRF-8 + WSPS-8 measure phase ACLR specifications, refer to the section "Wave Scale RF-8 measure ACLR (Topic 354351)". The specifications are also repeated below for completeness.

WSRF-8 + WSPS-8 measure ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC, Center frequency: 2 GHz

Power level	ACLR [dBc]
-16 dBm to +6 dBm	-67.0 (max)
-16 dBm to +6 dBm	-69.0 (typ)

Wave Scale PS-8 measure spurs

Measured at 0 dBm stimulus power

For the combined WSRF-8 + WSPS-8 measure spur specifications, refer to the section "Wave Scale RF-8 measure spurs (Topic 354590)". The specifications are also repeated below for completeness.

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF + WSPS measure spurs (measured values)

Input frequency from [GHz]	Input frequency to [GHz]	Max spurs [dBc]
0.13	4	-70

Input frequency from [GHz]	Input frequency to [GHz]	Max spurs [dBc]
4	6	-68
6	8	-67

Wave Scale PS card, using N2601PD direct pogo cable

The Wave Scale Precision Source card using N2601PD direct pogo cable (WSPS-PD or WSPS-PD-8) connects directly to the DUT-IF to provide high precision converter clocks and ADC signals. One WSPS-PD/WSPS-PD-8 card provides three independent high precision clock sources and fan-out up to four outputs per clock source, resulting in 12 outputs at the DUT interface level.

For information on the WSPS card which is attached to the WSRF card using N2601P interconnect cable, refer to the section "Wave Scale Precision Source card, using N2601P interconnect cable (Topic 255027)".

Wave Scale PS-PD phase noise

WSPS-PD phase noise max [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-139	-149	-152	-152	-152
320	-119	-136	-138	-149	-150
545	-117	-134	-138	-149	-149
600	-117	-134	-138	-149	-149
885	-114	-131	-133	-147	-148
1900	-109	-125	-127	-145	-145
2400	-106	-124	-126	-144	-145
5800	-95	-118	-118	-143	-147

WSPS-PD phase noise typ [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-145	-154	-156	-156	-156
320	-129	-143	-149	-155	-155
545	-125	-139	-143	-153	-153
600	-124	-138	-143	-153	-153
885	-121	-135	-138	-151	-151
1900	-115	-130	-132	-148	-149
2400	-112	-127	-131	-148	-148
5800	-104	-122	-124	-148	-151

Wave Scale PS-PD random jitter

WSPS-PD random jitter max

Frequency [MHz]	Random jitter max [$f_{s_{rms}}$]
600 $\leq f < 800$	37
800 $\leq f < 1000$	34
1000 $\leq f < 1900$	28
1900 $\leq f < 2400$	24
2400 $\leq f < 4000$	24
4000 $\leq f < 5000$	21
5000 $\leq f \leq 6000$	17

WSPS-PD random jitter typ

Frequency [MHz]	Random jitter typ [$f_{s_{rms}}$]
600	28
885	27
1900	20
2400	17
5800	12

Wave Scale PS-PD output power

WSPS-PD typical output power

Frequency [GHz]	Output power, typical [dBm]
0.5	-1.0 $\pm$ 0.3 dB
1	-1.1 $\pm$ 0.3 dB
3	-1.7 $\pm$ 0.4 dB
6	-3.3 $\pm$ 0.6 dB

Wave Scale PS-PD-8 phase noise

WSPS-PD-8 phase noise max [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-139	-149	-152	-152	-152
320	-119	-134	-138	-149	-150
545	-117	-134	-138	-149	-149

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
600	-117	-134	-138	-149	-149
885	-114	-130	-133	-147	-148
1900	-109	-124	-127	-145	-145
2400	-106	-122	-126	-144	-145
5800	-95	-116	-118	-142	-147
7000	-95	-113	-116	-137	-143

WSPS-PD-8 phase noise typ [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-146	-154	-156	-157	-157
320	-130	-145	-149	-157	-157
545	-126	-140	-144	-156	-156
600	-124	-139	-144	-156	-156
885	-121	-134	-138	-153	-154
1900	-115	-129	-132	-150	-151
2400	-112	-128	-132	-150	-151
5800	-105	-121	-125	-147	-150
7000	-103	-117	-120	-142	-147

Wave Scale PS-PD-8 random jitter

WSPS-PD-8 random jitter max

Frequency [MHz]	Random jitter max [$f_{s_{rms}}$]
600 $\leq f < 800$	37
800 $\leq f < 1000$	34
1000 $\leq f < 1900$	28
1900 $\leq f \leq 2400$	24
2400 $< f < 4000$	24
4000 $\leq f < 5000$	21
5000 $\leq f \leq 6000$	17
6000 $< f \leq 8000$	25

WSPS-PD-8 random jitter typ

Frequency [MHz]	Random jitter typ [$f_{s_{rms}}$]
600	22

Frequency [MHz]	Random jitter typ [$f_{s_{rms}}$]
885	23
1900	17
2400	18
5800	11
7000	16

Wave Scale PS-PD-8 output power

WSPS-PD-8 typical output power

Frequency [GHz]	Output power, typical [dBm]
0.5	0.9 ± 0.2 dB
1	0.8 ± 0.2 dB
3	0.2 ± 0.2 dB
6	-1.1 ± 0.4 dB
7	-1.8 ± 0.5 dB
8	-2.8 ± 0.5 dB